LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Components of rotor mast, control system, swash plate, torque link, drag damper, droop restrainer, clutch mechanism, rotor brake, feathering
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Components of rotor mast, control system, swash plate, torque link, drag damper, droop restrainer, clutch mechanism, rotor brake, feathering sleeve etc.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of rotor mast components and control linkages on a cutaway model; Safety: Wear eye protection; Instructor: Supervise handling of mechanical parts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the main rotor mast?
(b) How does the swash plate convert pilot input into blade pitch changes?
(c) Why is the droop restrainer critical during engine failure?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Rotor mast structural components • Control system linkage arrangement • Swash plate pitch and tilt mechanics • Torque link function and drag damper • Droop restrainer operation principles	L & Ds	PPT, IB	15 min
(b)	• Clutch mechanism engagement logic • Rotor brake application procedures • Feathering sleeve operation details • Vibration check procedures • Power and auto rotation controls	L & Ds	PPT, PDF	20 min
(c)	• Translation flight vector analysis • Main rotor movement dynamics • System adjustments and maintenance • Control system troubleshooting basics • Integration of rotor and control systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the torque link in the control system.
(b) Describe the sequence of events when the rotor brake is applied.
(c) How does the feathering sleeve assist in auto-rotation?

12. Home Assignment and Follow Up	Prepare a brief note comparing the control systems of a single-rotor helicopter with a contra-rotating rotor system, referencing the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Control horizontal and vertical movement, control of RPM, torque reaction, tail rotor compensations, contra rotating rotors.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min